
Research Proposal

Deep Interpretable Models for Clinical Applications

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ABSTRACT

Deep Neural Networks (DNNs) have shown practical success in a number of tasks in the medical domain, such as patient deterioration detection using EHR data, brain-tumor segmentation, and diabetic retinopathy classification. As such techniques become deeply integrated into medical diagnosis, it is necessary to make sure they are transparent and accessible to medical professionals. While the need for interpretability in the clinical context has been well emphasized, there is a limited amount of work on the development and application of interpretable deep learning techniques in this area. In this proposal, I have explored potential directions for evaluating and building such interpretable deep models for clinical applications. While this proposal focuses on interpretability, I am also interested in other research directions, like generalization, privacy, and deployment on edge devices, and the transferability of these properties across domains.

1 OBJECTIVES & SCOPE

In the medical domain, deep learning has achieved human-like or even better results on many problems. Interpreting the decisions of such models is especially important here, where transparency is essential from a regulatory point of view and to make sure that predictions can be trusted by medical professionals. The aims of this research proposal are two-fold - 1) to evaluate existing interpretability approaches from the context of clinical applications, and 2) to explore novel interpretability techniques in conjunction with medical professionals which may make specific sense for this domain.

1.1 Evaluating existing interpretability approaches

There has been a large body of work on interpretable deep learning techniques in the past half decade (Molnar, 2020). The need for interpretability in a clinical context has been emphasized before (Ahmad et al., 2018), and will not be reviewed here.

Ahmad et al. (2018) emphasise that interpretability in the medical domain is context dependent. This part of project will aim to evaluate and benchmark existing interpretability techniques in the clinical context, i.e. try to understand which method is most accessible to medical professionals and users for a particular task. I borrow from Gilpin et al. (2018) to broadly divide existing interpretability techniques into: (1) Processing based methods, like linear proxy models and rule extraction, (2) Attribution based methods, like saliency or gradient-backpropagation, (3) Representation based methods, which elucidate the role of different parts of the model, and (4) Explanation producing systems, like conversation agents or sample-based explanations.

Depending on the nature of the data, different interpretability techniques may be useful for a medical practitioner. For example, CNNs are usually explained using attribution or representation based methods.

RNN interpretability methods use attention based mechanisms to highlight variables and time-steps which may be important to predictions (Shickel et al., 2019).

However, there has been no study (to the best of my knowledge) to understand and evaluate the suitability of interpretability of deep learning approaches for specific problems in the medical domain. For example, which class of methods as divided above would be most useful to clinical technicians while using a deep RNN for assessing patient mortality in the ICU? Which type of interpretability methods would provide most value to radiologist when using deep models for brain-tumor segmentation? The aim of this section is to answer such questions by creating a framework for evaluating and benchmarking interpretability techniques in conjunction with medical professionals. This would mean trying to quantify the value added by a potential interpretability technique over and above the predictions of the model itself. This would also be the first step towards developing novel interpretability techniques which best fit the clinical context, which is discussed in the next section.

1.2 Exploring novel interpretability techniques

Once the limitations of current interpretability techniques have been evaluated, we can move towards developing new techniques by actively integrating feedback from medical professionals. Two possible drawbacks of current interpretability approaches are: (1) They are post-hoc, i.e. they try to extract useful explanations from already trained deep models without including medical professionals in the training process, and (2) They do not provide concept based interpretability. Developing new techniques of interpretability will aim at addressing these as well as other issues that come up from the initial part of the project.

Concept-based interpretability intends to mimic the concept-based thinking that human-beings show. Providing interpretations in the form of human understandable concepts instead of at a variable or sample level would be especially useful in the medical domain. In our recent work (Kori et al., 2020) on interpreting brain-tumor and diabetic retinopathy models, we show that human-understandable concept-based explanations can be extracted organically from the model itself without manually creating concepts in the input space. In the context of time series data, the structure of RNN models can be changed to provide attention over groups of input variables or 'concepts' instead of individual variables.

Moving towards ante-hoc interpretability would involve integrating knowledge from medical professionals in the training process itself to increase intrinsic interpretability of models. This could be done by providing additional domain knowledge to models while training, apart from the data they are trained on, moving from black-box models to an interactive ML or glass-box approach (Holzinger et al., 2017).

In the next section, I detail two potential proposals for novel interpretability techniques, one for time-series EHR data and another for medical image data.

2 METHODS & DETAILED PROPOSALS

2.1 Concept Fusion for EHR data

Broadly, this involves using a learnt or user-defined hierarchical attention mechanism for grouping raw EHR features into abstract, high level concepts. For instance, if medical disease codes were taken as an example, in the ICD9 code hierarchy¹, the concept 'Diseases of the circulatory system' is an umbrella concept that branches into more specific concepts. For medical disease code, this hierarchy is explicitly available, but it may not be for other EHR data. Given a set of EHR features, such as medical disease

¹ <http://www.icd9data.com/2015/Volume1/default.htm>

codes, bedside monitoring data, or medical tests conducted, the aim would be to either manually create or to automatically learn higher level concepts from these low-level features. As in Choi et al. (2017), these multi-level features can then be combined and fed to the RNN model for prediction. Integrating attention over these multi-level features would then provide higher level concept-interpretability as opposed to simple attention on features and time-steps.

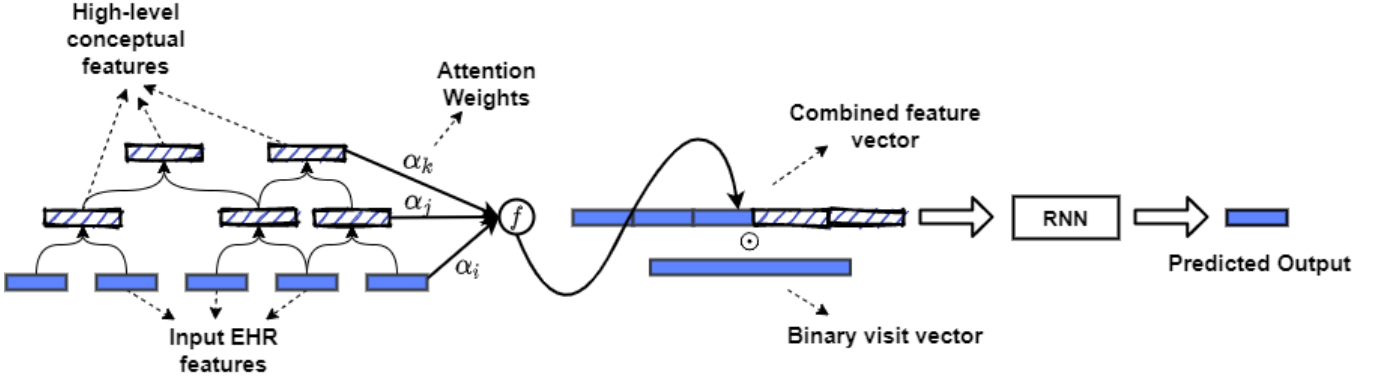


Figure 1. The proposed pipeline for concept attention based interpretability of RNN models for EHR data. The pipeline is shown on a single time-step for simplicity.

Mathematically, assume that $\mathcal{F} = (f_1, f_2, \dots, f_n)$ is the set of raw input EHR features and $\mathcal{V}_t \in \{0, 1\}^{\mathbb{R}^n}$ is the binary visit vector at timestep t for a single patient where each v_t^i indicates whether an EHR feature f_i was detected at that timestep. Let $\mathcal{F}' = (f'_1, f'_2, \dots, f'_m)$ be the higher level concepts generated from the EHR features. Then,

$$\mathcal{F}' = \phi(\mathcal{F}) \quad (1)$$

$$\mathcal{C} = f(\alpha_i f_i, \alpha'_i f'_i) \quad (2)$$

$$x_t = g(\mathcal{V}_t, \mathcal{C}) \quad (3)$$

$$o_{t+1} = RNN(x_t) \quad (4)$$

Where, ϕ is the function used to combine raw EHR features into higher level concepts, $\mathcal{C}$ are the feature embeddings after combining attention over the low and high level features using attention weights α_i , x_t is the input to the RNN for one timestep, and o_{t+1} is the prediction at timestep $t + 1$. Figure 1 shows this succinctly.

The process of generation of high-level conceptual features, which are context-specific, would require establishing co-occurrence and precedence between features and related concepts to create a hierarchy, and is another potential direction for research. This also provides the opportunity to move from post-hoc to ante-hoc interpretability by involving medical professionals in the concept or hierarchy generation process.

2.2 Directed Attention for Image Data

In previous work (Natekar et al., 2019), we show that deep learning models for brain tumor segmentation learn anatomically relevant disentangled concepts. There has been extensive research on directing model attention to anatomically relevant concepts. Cai et al. (2018) use visual saliency maps generated from

sonographer eye movements to augment CNNs for fetal ultrasound plane detection. Fan et al. (2020) use a self attention mechanism to localize regions in microscopic images, while Guan et al. (2018) use directed attention to enhance thorax disease classification. From the perspective of ante-hoc interpretability, model attention can be directed to anatomically relevant regions by using visual saliency maps annotated by medical professionals.

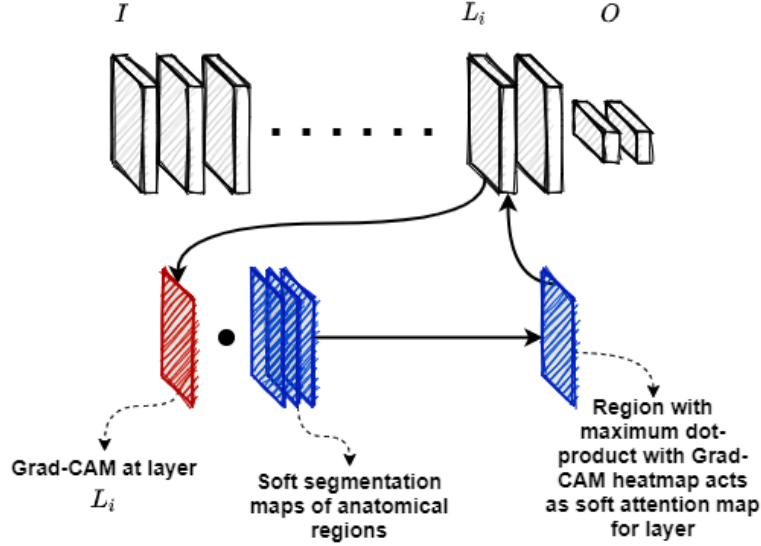


Figure 2. A graphical representation of the directed attention framework using soft anatomical segmentation maps.

Consider a CNN ϕ with layers $L_1, \dots, L_n$. Let $M_1, M_2, \dots, M_k$ be a set of soft segmentation masks for different anatomical regions. First, the CNN is trained for N iterations (which is a predetermined hyperparameter). We then find the Grad-CAM attention map for each layer L_i . Given our Grad-CAM attention maps, we can find the anatomical region with the highest overlap with the attention map by finding the maximum of the hadamard-product between the Grad-CAM for a layer G_i and the soft segmentation masks M_j . The layer is then multiplied by the soft segmentation map with the highest overlap to direct attention towards the closest medically relevant anatomical region. This is based on the presumption that after a certain number of iterations, layers in the model already look at certain input regions, and we only need to direct them to the closest anatomically relevant concept. In recent work, we also show that learning *consistent* representations helps models generalize better (I recently won the NeurIPS 2020 competition on predicting Generalization in Deep Learning, and my work will be presented at NeurIPS this year). Such a directed attention framework would also encourage models to focus on consistent input concepts. Mathematically,

$$\mathcal{G}_{L_i} = \text{gradcam}(\mathcal{L}_i) \quad (5)$$

$$\mathcal{H}_{L_i} = \arg \max_{M_i} (\mathcal{G}_{L_i} \odot M_i) \quad (6)$$

$$\mathcal{A}_{k,l} = H_{L_i} \odot A_{k,l} \quad (7)$$

Where, H_{L_i} is the attention heatmap for layer L_i as determined above, and $A_{k,l}$ is the k^{th} feature map of layer L_i . We can then extract a clear hierarchy of inference steps by taking the Grad-CAM of the trained

network for each layer for a particular image, and see the attention over the relevant anatomical region the layer has been encouraged to focus on. Such a trace of anatomically relevant inference steps would help increase transparency and trust and allow us to move towards a glass-box approach. Figure 2 shows this process graphically.

3 TIMELINE

The Gantt Chart below shows an approximate timeline of the proposed project.

	Q1 Y1	Q2 Y1	Q3 Y1	Q4 Y1	Q1 Y2	Q2 Y2	Q3 Y2	Q4 Y2
Literature Review and Project Outline								
Benchmarking Existing Interpretability Techniques on Relevant Datasets								
Interaction with Medical Professionals to Understand Drawbacks, Advantages of Existing Techniques								
Developing New Interpretability Techniques Based on Requirements Derived from Above								
Evaluation and Comparison of New Techniques with Existing Methods; Quantification of Value Added								

Figure 3. Proposed Project Timeline. Q: Quarter, Y: Year.

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